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ONE FRAME. TWO TRUTHS.

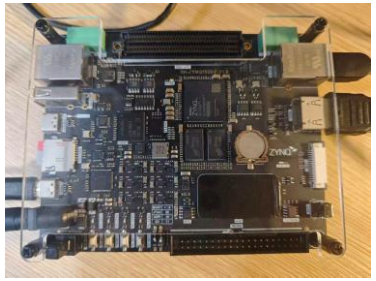
A full-hardware video node on Zynq-7020

多源视频流多效果变换处理系统

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湖南工业大学
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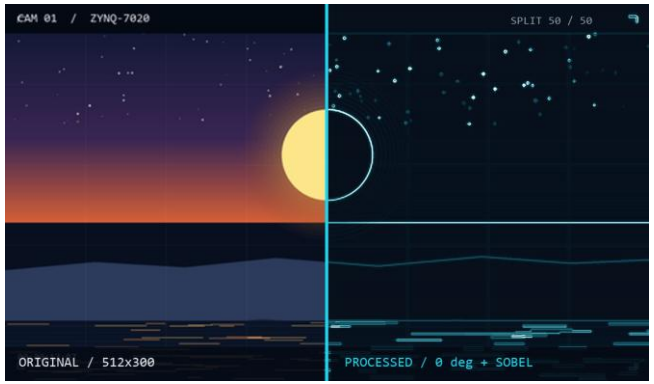
AMD
ZYNQ-7020



RK-ZYNQ7020-F V1.0
xc7z020clg484-2 · 1 GbE

INTRODUCTION

Remote video nodes usually have one Ethernet cable and no PC. The operator must know whether the frame arrived, whether processing took effect, and how old the picture is. This node answers all three from one frame: 512x300 RGB565 enters RGMII, is reassembled and committed to DDR only after three gates, then read back once for both panes. The left pane is the original; the right pane is the same frame after geometry and a fixed 15-cycle effect chain. The seam is a per-pixel decision, not two screenshots.

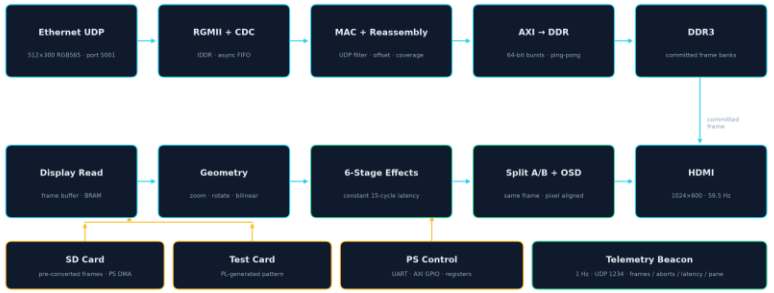


Same 512x300 frame · seam 50/50 · right pane 0 deg + Sobel

Key technical issues addressed:

1. 1 GbE UDP: offset reassembly; commit = bytes + 300 rows + no bad packet.
2. Same-frame A/B: one coordinate stream, one row map, 15-cycle chain.
3. Telemetry: aborts, rows_max, gaps, pane_bad; drop 1/100 reddens it.

SYSTEM ARCHITECTURE



RGMII -> UDP -> DDR ping-pong -> geometry -> effects -> A/B seam -> HDMI

THREE-GATE COMMIT

A frame is committed only when 307200 bytes arrived, all 300 source rows were written, and no packet failed. Otherwise the last good frame stays on screen and aborts + rows_max expose the reason.

SAME-FRAME A/B

Original and processed panes share one source-coordinate stream, one row map and one 15-cycle effect chain. The seam is a per-pixel select; rotation or stacked effects cannot make the panes disagree.

RESULT

RGMII -> HDMI entirely in PL; PS only control, SD playback and telemetry. 1024x600 at 59.5 Hz, 1 GbE UDP ingest.

Setup WNS	1.341 ns · 0 failing endpoints
Hold WHS	0.025 ns · positive margin
BRAM	70.0% · 98 / 140 tiles
8 s stream	241 frames · 29.95 fps
UART battery	105 commands · PASS
QSPI cold boot	DONE / EOS · no CRC error
Fault injection	22.0 / 0 / 18.9 fps · PASS

OBSERVABILITY

link / 1 Hz UDP telemetry: aborts, rows_max, stall, gap_min/max and pane_bad. The same counters are readable over UART, JTAG and the wire.

DROP 1 IN 100 -> ABORTS RISE -> RECOVERY

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